

Kelvin-Helmholtz instability

Kelvin-Helmholtz instability happens when there's shear in the contact discontinuity between two fluids.

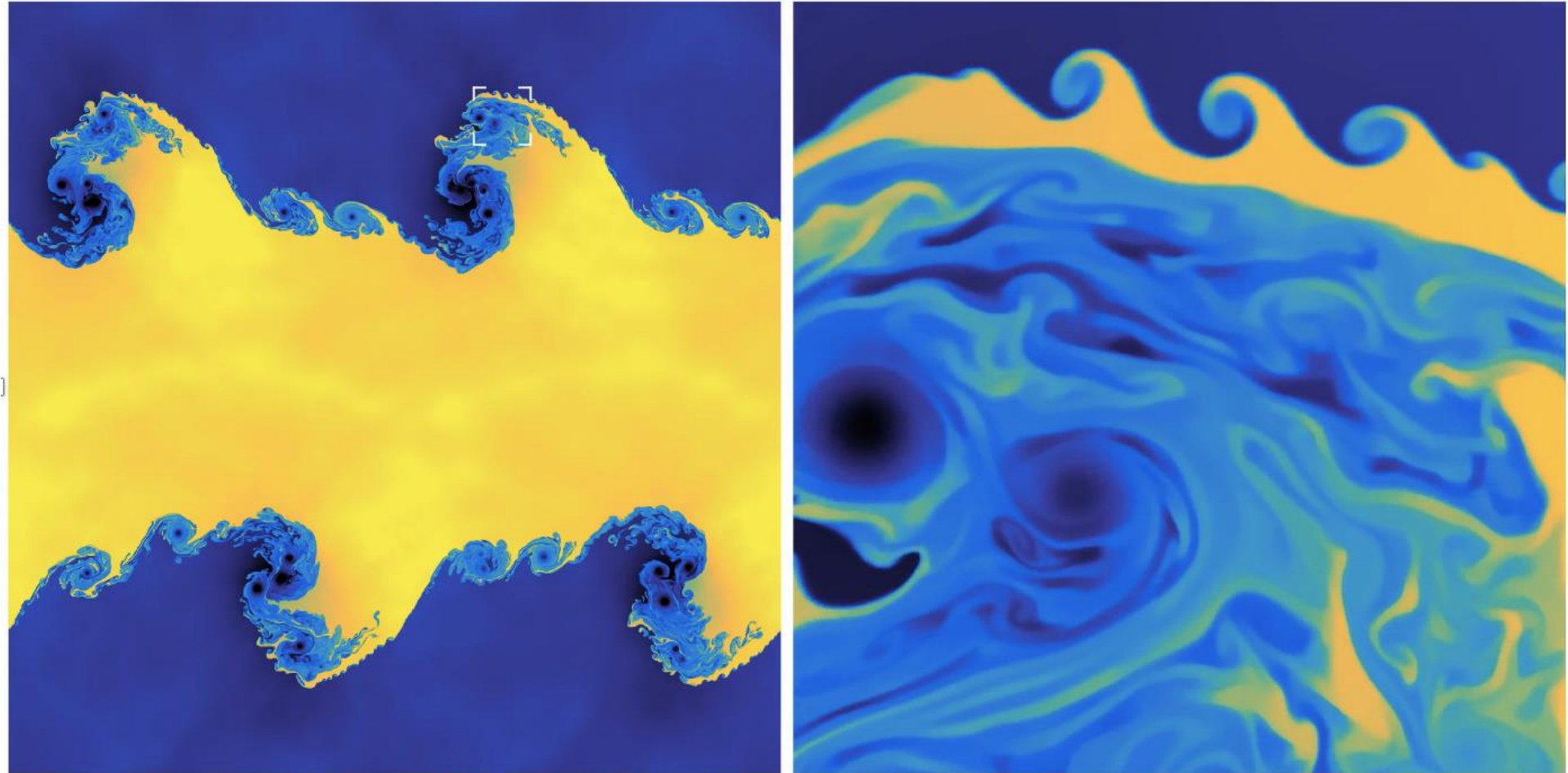
Here the two fluids are the atmosphere and the cloud deck.



Kelvin-Helmholtz instability

Simulation of
Kelvin-Helmholtz
instability.

Notice the fractal
character of the
instability.

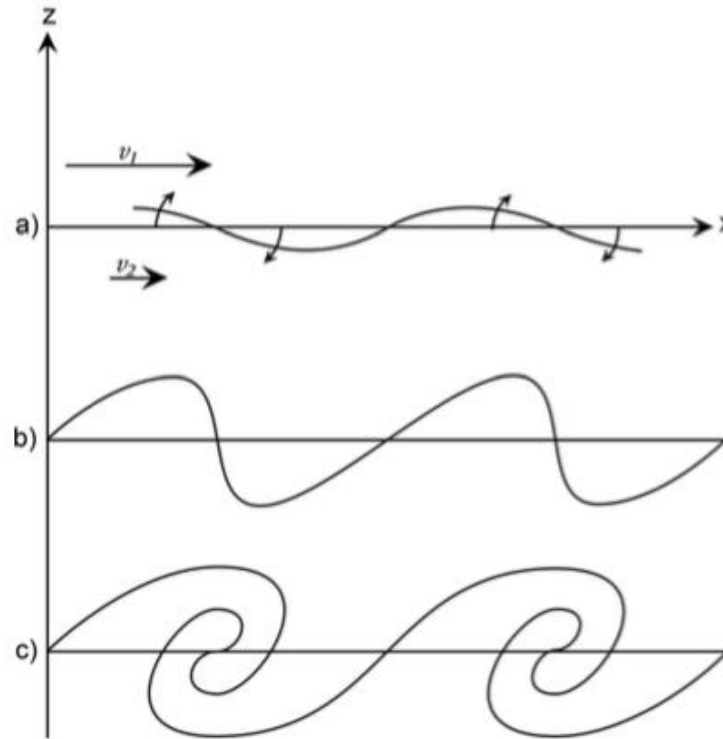


Physical interpretation

A wavefront of a mode between two fluids will be stretched by the shear.

Gradients steepen and the mode will be twisted into a billow: the instability converts shear into vorticity.

Kelvin-Helmholtz Instability



Geometry of Kelvin-Helmholtz instability in accretion disks

Contact discontinuity between fluids 1 and 2,
shear between them, no gravity.

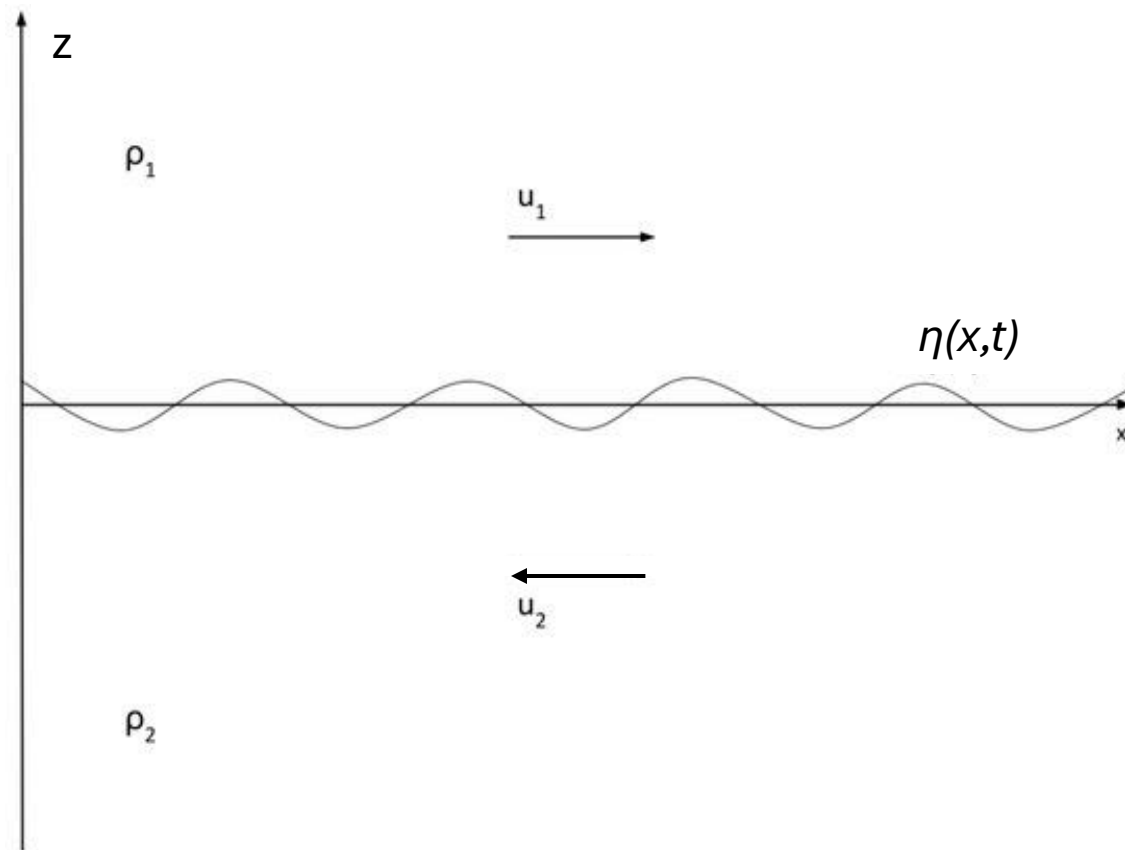
$\eta(x,t)$ is a perturbation of the interface.

The question we ask is: what is the oscillation
frequency of $\eta(x,t)$?

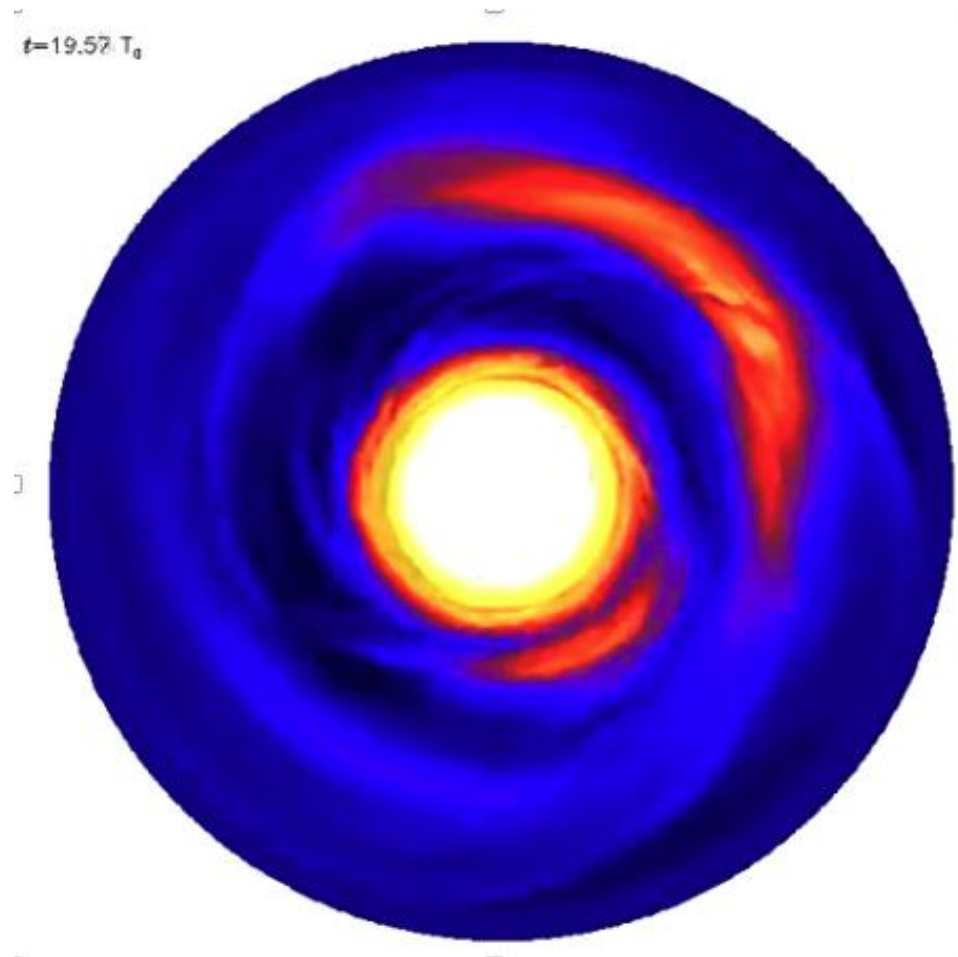
The solution is the dispersion relation

$$\omega_{1,2} = \frac{k(\rho_1 u_1 + \rho_2 u_2)}{\rho_1 + \rho_2} \pm i \frac{\sqrt{\rho_1 \rho_2}}{\rho_1 + \rho_2} |u_1 - u_2|$$

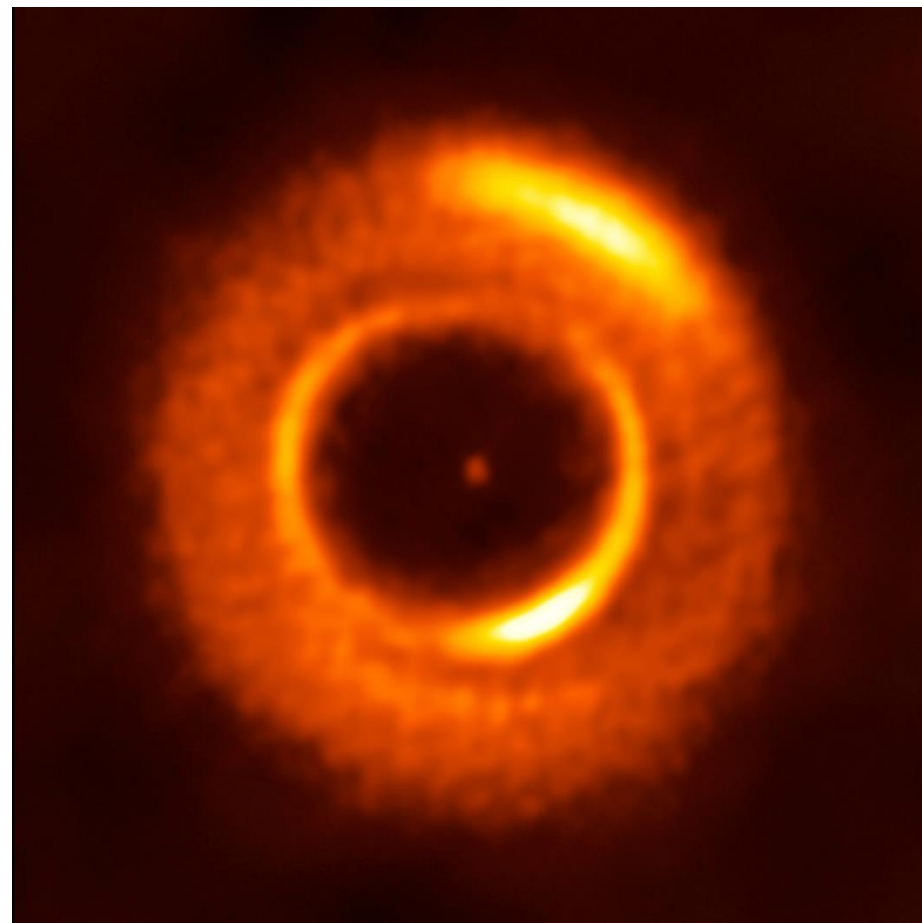
Exponential growth (instability) exists when $|u_1 - u_2| > 0$



Example: Kelvin-Helmholtz instability in accretion disks



Model
Lyra & Mac Low (2012)



Observation – MWC 758
Dong et al. (2018)

Example:

Measurements of Kelvin-Helmholtz Waves in Earth's Magnetic Field

Simultaneous satellite observations from different distances of Earth's magnetic tail offer insight into how these instability waves evolve through time and space.

By Terri Cook 20 July 2018



This artist rendering shows how Kelvin-Helmholtz instability waves in Earth's magnetic tail might propagate through time and space. Credit: Quanqi Shi

Example:

nature

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nature > letters > article

Letter | Published: 12 August 2004

Transport of solar wind into Earth's magnetosphere through rolled-up Kelvin–Helmholtz vortices

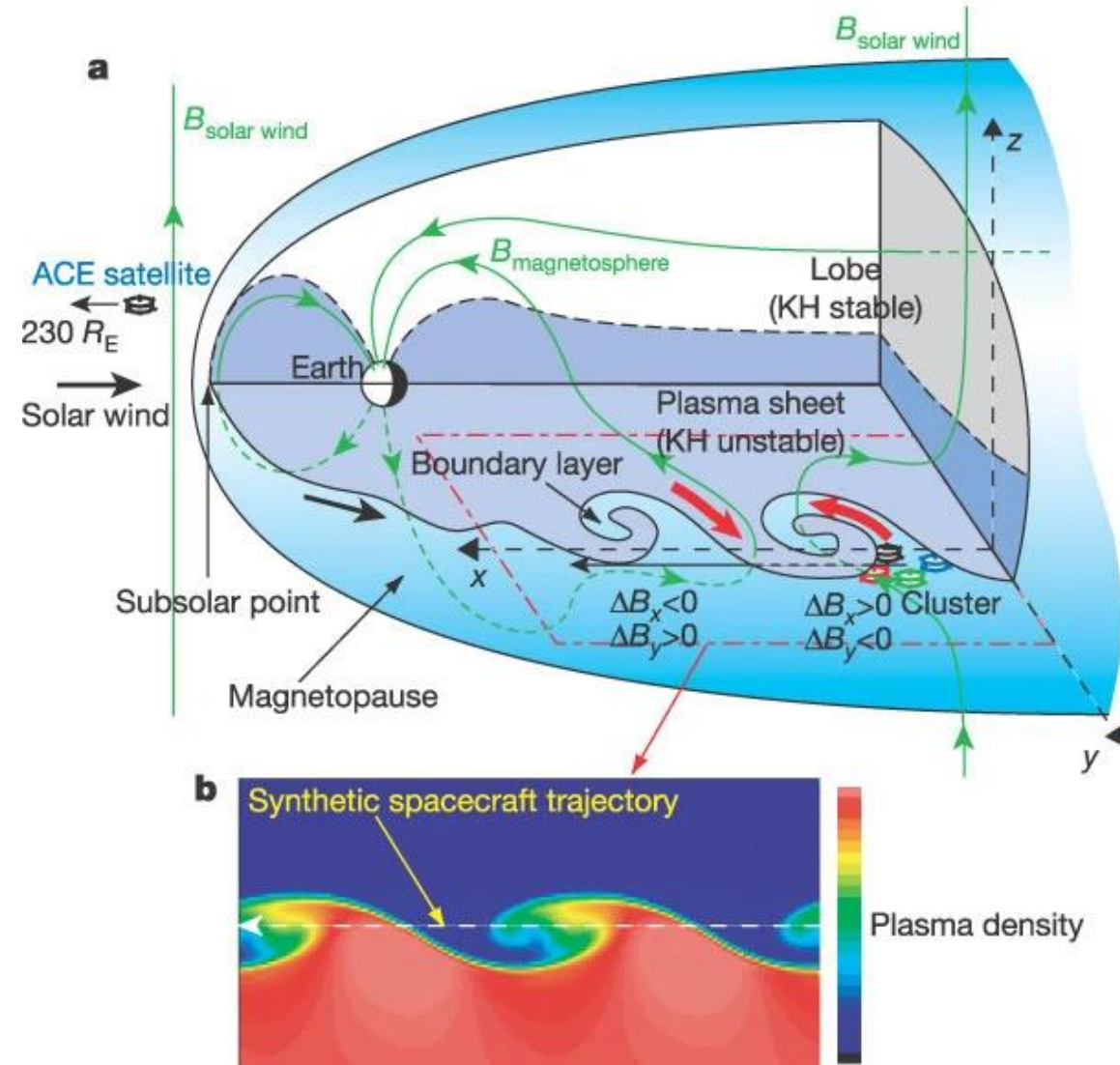
H. Hasegawa , M. Fujimoto, T.-D. Phan, H. Rème, A. Balogh, M. W. Dunlop, C. Hashimoto & R. TanDokoro

Nature 430, 755–758 (2004) | [Cite this article](#)

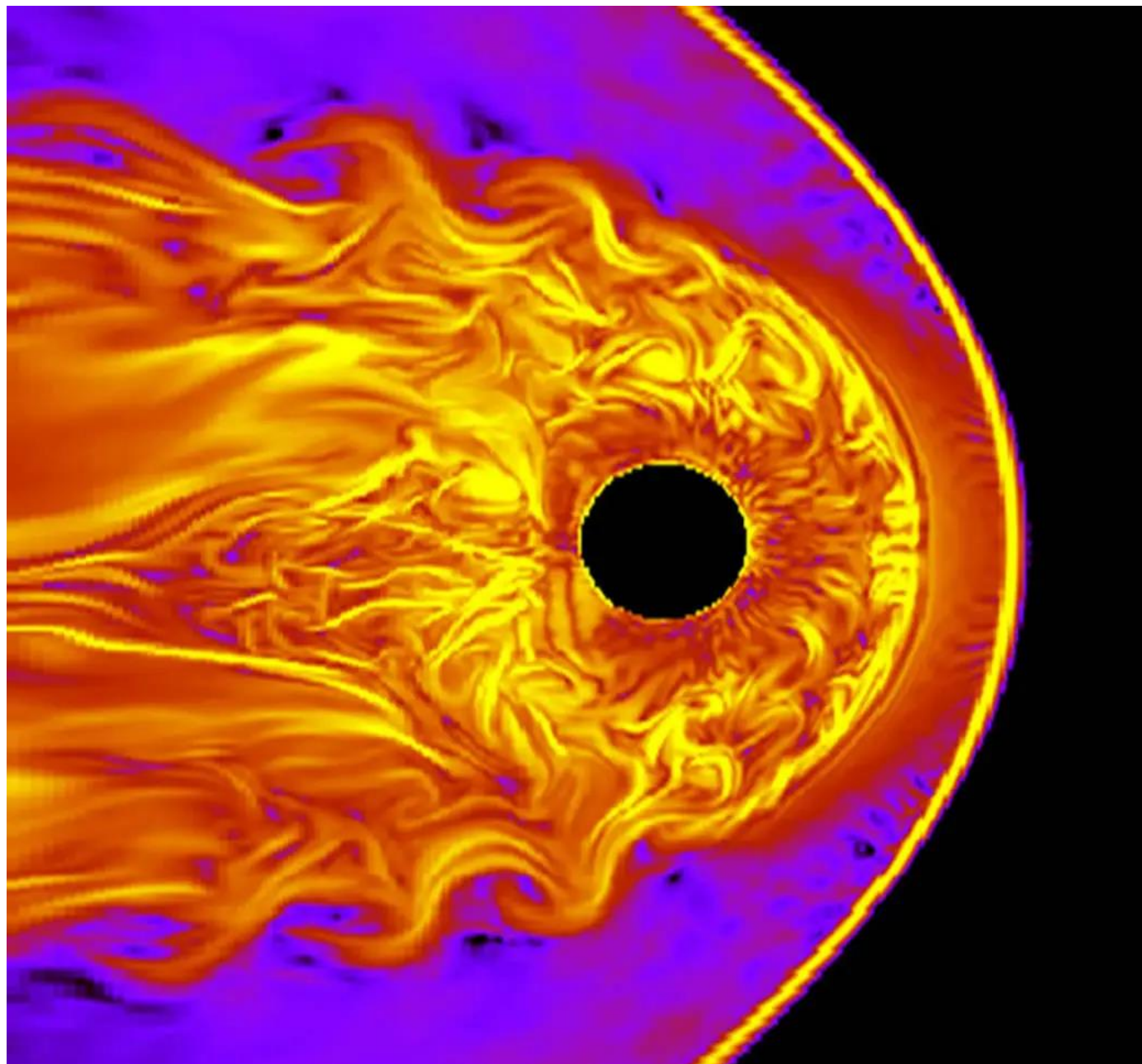
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Abstract

Establishing the mechanisms by which the solar wind enters Earth's magnetosphere is one of the biggest goals of magnetospheric physics, as it forms the basis of space weather phenomena such as magnetic storms and aurorae¹. It is generally believed that magnetic reconnection is the dominant process, especially during southward solar-wind magnetic field conditions when the solar-wind and geomagnetic fields are antiparallel at the low-latitude magnetopause². But the plasma content in the outer magnetosphere increases during northward solar-wind magnetic field conditions^{3,4}, contrary to expectation if reconnection is dominant. Here we show that during northward solar-wind magnetic field conditions—in the absence of active reconnection at low latitudes—there is a solar-wind transport mechanism associated with the nonlinear phase of the Kelvin–Helmholtz instability⁵. This can supply plasma sources for various space weather phenomena.



Example:

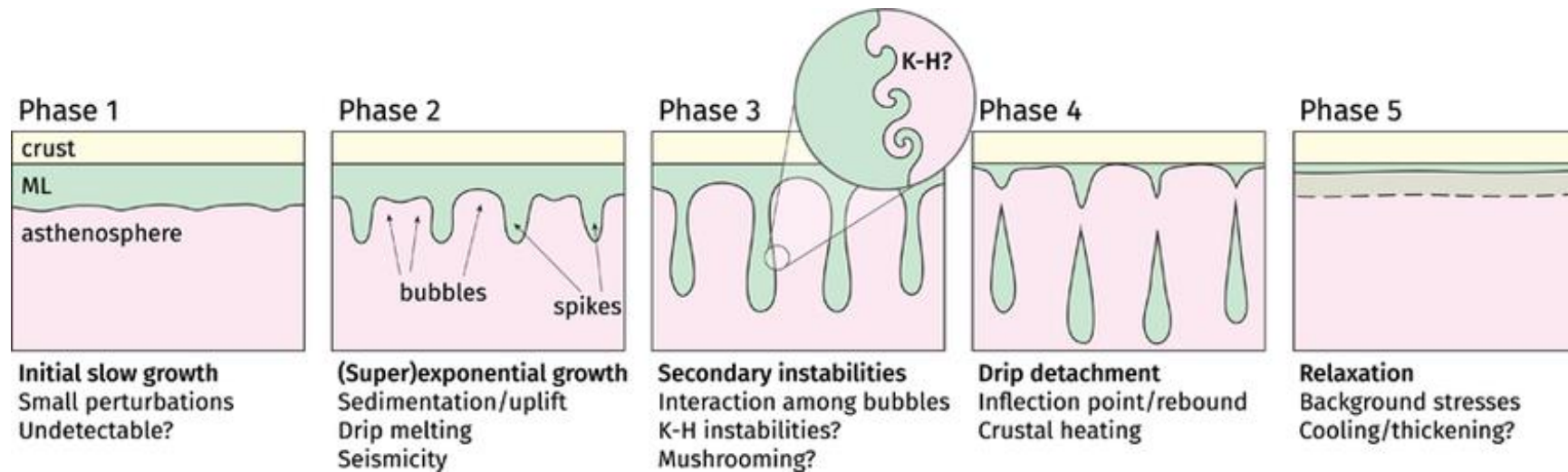


Example:



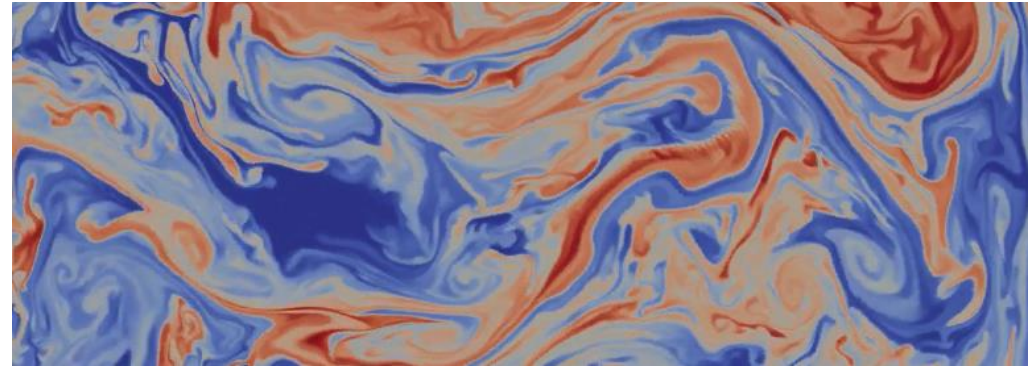
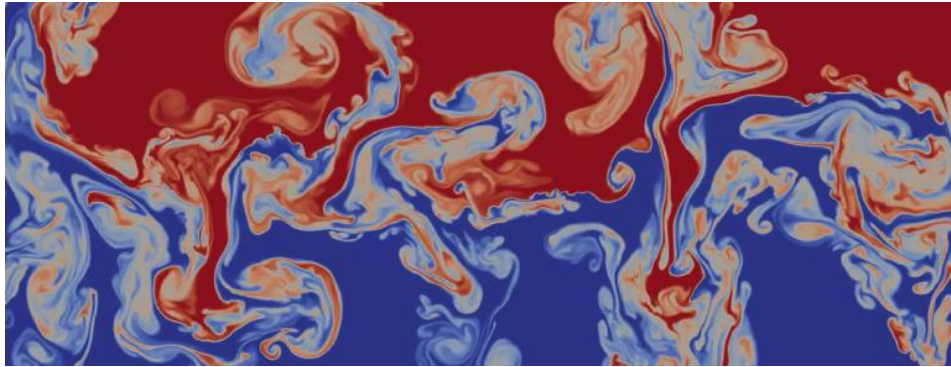
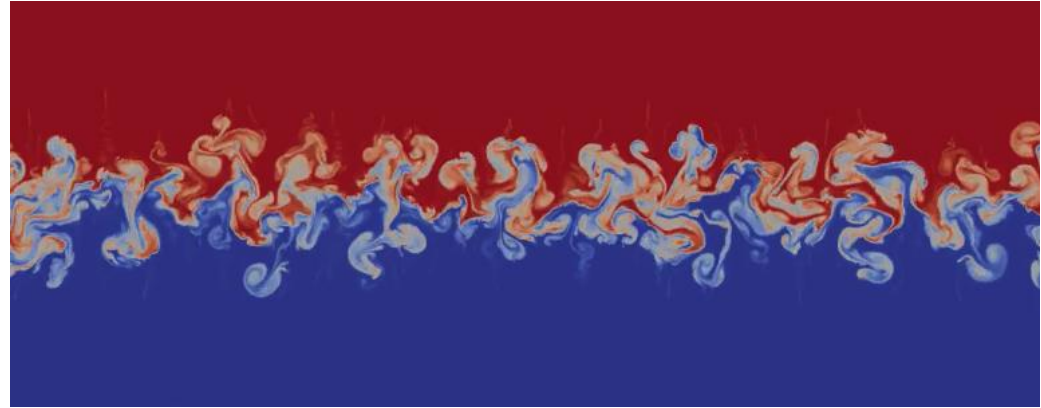
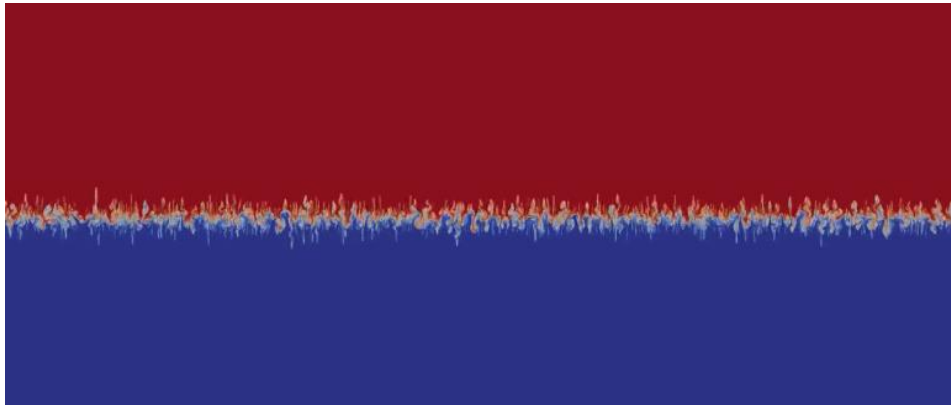
Transition to turbulence

Once the Rayleigh-Taylor instability develops, it builds shear between the fluids. This shear is unstable to Kelvin-Helmholtz, leading to vortices, that in turn will also become Kelvin-Helmholtz unstable.



Transition to Turbulence

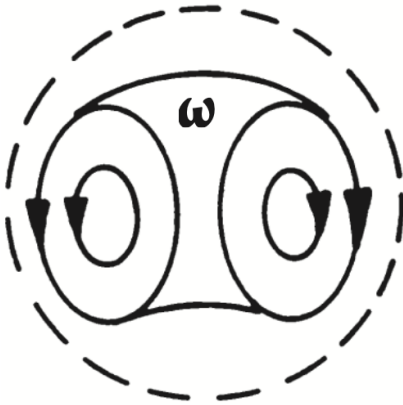
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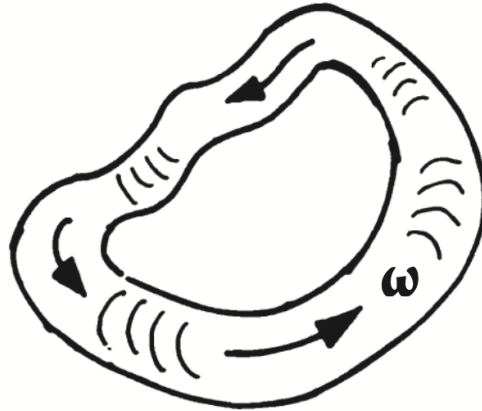
Transition to turbulence

Typical vortex shapes in turbulence

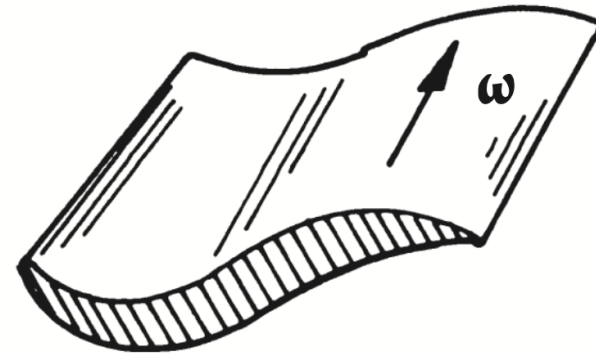
Blobs



Tubes

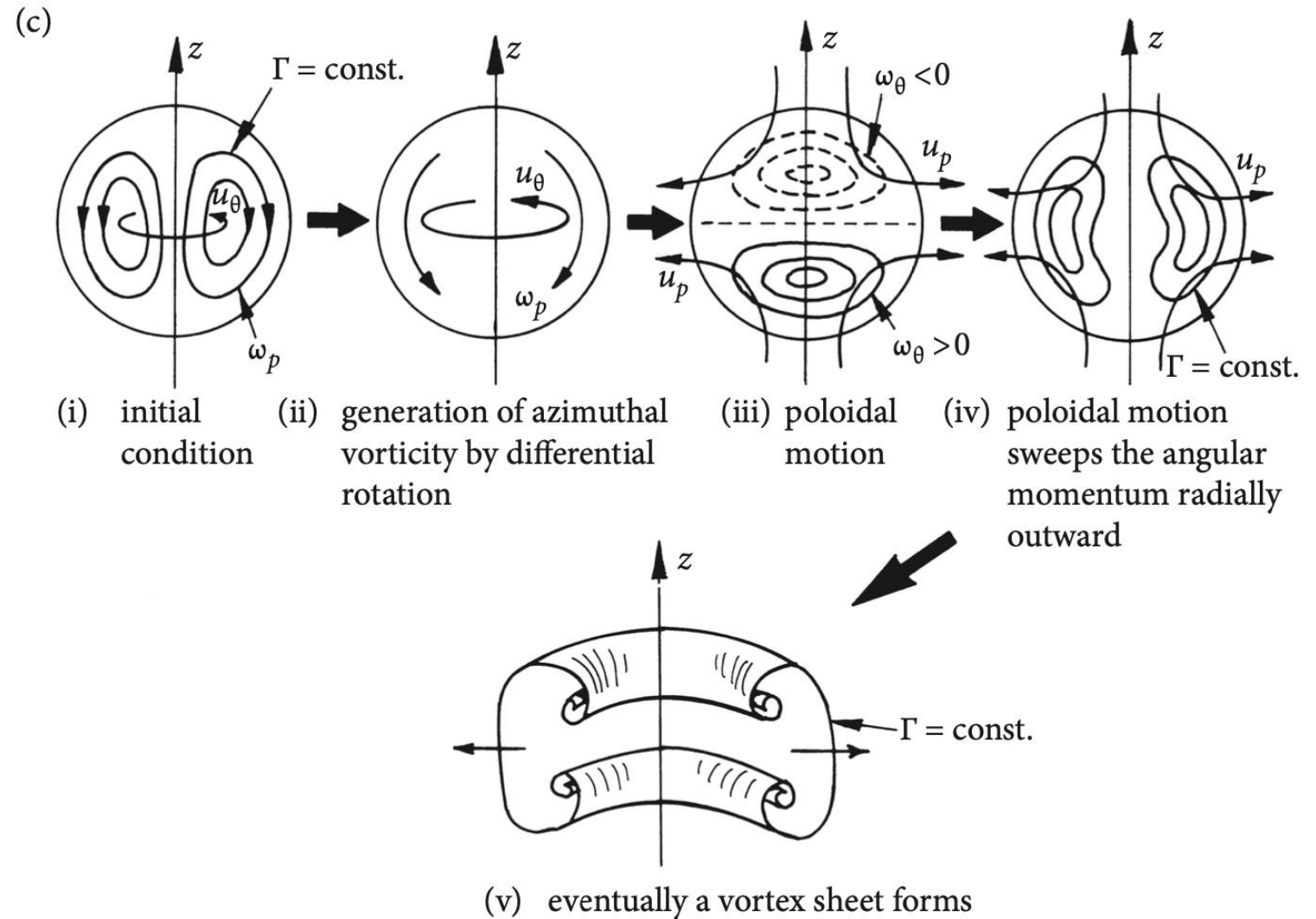


Sheets

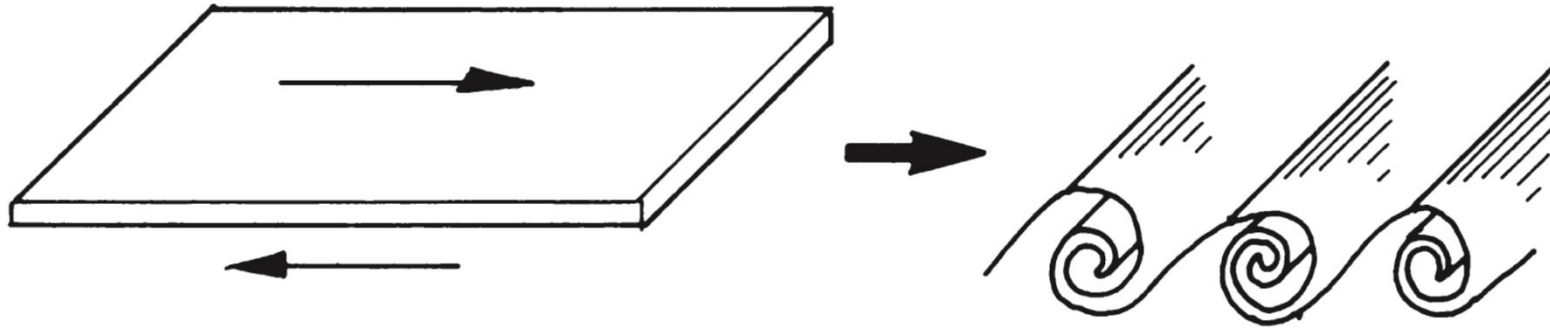


Bursting of vortex blobs

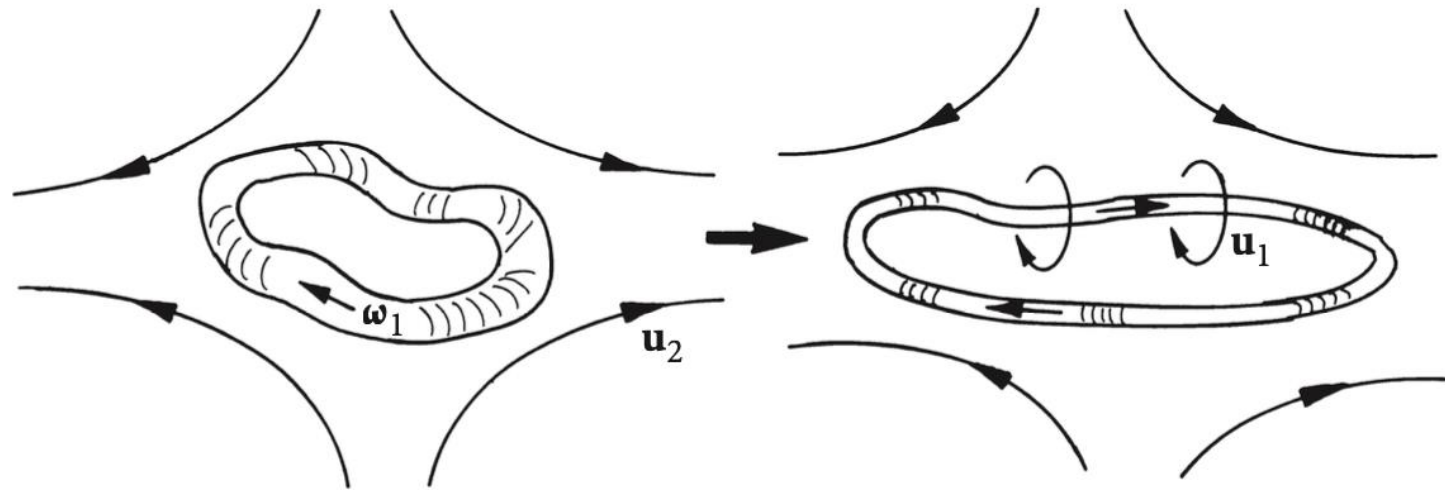
Vortex blobs turn into sheets



**Vortex sheets roll up into vortex tubes
due to Kelvin-Helmholtz instability**



Vortex tubes stretch in the strain field of other vortices



Summary: Blobs burst into sheets, sheets roll up via KH into tubes, tubes get stretched thin. The end result is the cascade, i.e., the transfer of energy to smaller scales.